

Dhirubhai Ambani University
Technology
Formerly DA-IICT

MediNexus Healthcare Ecosystem

Database Project Scenario Description

Course	IT214 - Database Management Systems
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Project Team

Name	Roll Number
Atharv Trivedi	202401230
Jay Vaghela	202401237
Dharmraj Vaghela	202401236
Shlok Topaliya	202401229

Group Representative: Shlok Topaliya
Contact No.: +91 9082919527

1 Objective

Design and implement a database for a comprehensive digital healthcare and telemedicine ecosystem. The platform, MediNexus, serves as a unified marketplace connecting patients with healthcare providers and pharmacies. It manages the entire lifecycle of medical care — from remote video consultations and Electronic Health Records (EHR) to automated pharmacy inventory management.

2 Scope of the Database System

The MediNexus database is designed to support a modern healthcare and telemedicine ecosystem. The system manages healthcare data related to patients, doctors, appointments, prescriptions, medical records, and telemedicine consultations. The database enables efficient storage, retrieval, and analysis of healthcare information to support hospital operations and remote healthcare services.

2.1 Included in Scope

The database system will manage and store information related to the following activities:

- Patient registration and profile management
- Doctor registration and specialization information
- Appointment scheduling for both physical and telemedicine consultations
- Maintenance of patient medical history and diagnosis records
- Prescription generation and medicine details
- Pharmacy inventory management
- Consultation billing and payment records
- Telemedicine consultation tracking
- Healthcare operational reports and analytics

2.2 Excluded from Scope

The following components are outside the scope of this database system:

- Real-time video consultation infrastructure
- Storage of large medical imaging files such as MRI or CT scans
- Integration with national healthcare systems

The focus of the database is to manage structured healthcare data and support the operational activities of a healthcare and telemedicine platform.

3 Application Users

User Type	Description
Surfer (Unlogged Users)	Users exploring healthcare services without registering
Patients (Registered Users)	Registered users managing health records and consultations
Doctors / Medical Practitioners	Healthcare professionals providing diagnosis and treatment
Pharmacy Organizations	Partner pharmacies managing medicines and prescriptions
MediNexus Platform Admin	Administrator responsible for overall platform operations

4 Use Cases

4.1 Unlogged Users (Surfers)

- Search doctors based on specialization, consultation fee, language, and hospital affiliation.
- Compare healthcare providers based on average patient ratings and years of experience.
- Browse medical specialties to understand available treatments and diagnostic services.
- Check medicine availability in partner pharmacies by searching generic or brand names.

4.2 Patients

- Register and Health Profile Management including blood group and emergency contacts.
- Appointment Booking for either virtual consultation or in-person clinic visits.
- Health Vault Access to view prescriptions, lab reports, and clinical summaries.
- Vital Logging including blood pressure, pulse and glucose levels.
- Feedback and Reviews for doctors and pharmacies.

4.3 Doctors / Practitioners

- Profile and Schedule Management defining consultation hours and fees.
- Clinical Consultations accessing patient history and recording diagnoses.
- E-Prescription Issuance generating digital prescriptions.
- Referral and Lab Orders directing patients to diagnostic centers.

4.4 Pharmacy Organizations

- Pharmacy Inventory Management to maintain drug stock levels and prescription fulfillment.

5 Data Requirements and Details

5.1 Registration and User Identity

All users receive a unique System ID upon registration.

- Pharmacies must register with a valid Tax Identifier and business address.

- Doctors must provide Medical License Number and specialization details.

5.2 Medical Records and Versioning (EHR)

- Each consultation records mode (Video or Physical), fee and status.
- Patients can record multiple vital logs including metric type, value and timestamp.
- Prescriptions include multiple medicines with dosage, duration and quantity.

5.3 Pharmacy and Supply Chain

- A medicine identified by a generic code can be sold by multiple pharmacies.
- The system tracks inventory levels of each medicine at each pharmacy branch.

5.4 Financials and Billing

- Every consultation generates a bill.
- Payment records include gateway name and transaction reference ID.

6 Tentative List of Reports (Queries)

6.1 For General Users

- Find top-rated cardiologists charging less than 1500 INR.
- Compare price of medicines across pharmacies.
- List doctors available for video consultation within the next 24 hours.
- Search for pharmacies within a specific PIN code currently open.

6.2 For Patients

- My Medical History
- Vital Trends
- Prescription Status
- Upcoming Appointments

6.3 For Doctors

- Daily Schedule
- High-Risk Patients
- Consultation Summary
- Referral Tracking
- Earnings Report

6.4 For Pharmacy Admins

- Low Stock Alert
- Sales Summary
- Order Fulfillment Log

6.5 For MediNexus Global Admin

- Specialty Trends
- Platform Earnings
- User Growth
- Feedback Analytics
- Billing Reconciliation